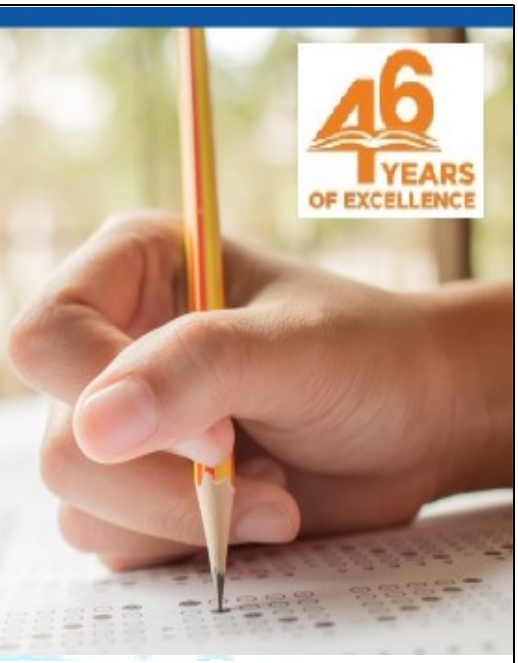
NARAYANA IIT ACADEMY INDIA

JEE ADVANCED GRAND TEST SERIES



46 YEARS OF EXCELLENCE

NARAYANA IIT - JEE ACADEMY - INDIA

Sec: OSR.IIT _*CO-SC(MODEL-A&APEX) **Date: 05-04-26**
Time: 3 Hrs **Max. Marks: 180**

05-04-26_OSR.IIT_*CO-SC(MODEL-A&APEX)_JEE ADV_GTA-3(P2)_SYLLABUS

Rankers Academy JEE

MATHEMATICS: **TOTAL SYLLABUS**

PHYSICS: **TOTAL SYLLABUS**

CHEMISTRY: **TOTAL SYLLABUS**

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TIME: 3HRS **IMPORTANT INSTRUCTIONS** Max Marks: 180

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 8)	Questions with Single digit integer(0-9)	+3	-1	8	24
Sec – II (Q.N : 9 – 14)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 15 – 18)	Single Correct option type	+3	-1	4	12
Total				**18**	**60**

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 19 – 26)	Questions with Single digit integer(0-9)	+3	-1	8	24
Sec – II (Q.N : 27 – 32)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 33 – 36)	Single Correct option type	+3	-1	4	12
Total				**18**	**60**

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 37 – 44)	Questions with Single digit integer(0-9)	+3	-1	8	24
Sec – II (Q.N : 45 – 50)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 51 – 54)	Single Correct option type	+3	-1	4	12
Total				**18**	**60**

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MATHEMATICS **MAX.MARKS: 60**

SECTION – 1 (Maximum Marks: 24)

This section contains SIX (08) questions.
The answer to each question is **A SINGLE DIGIT INTEGER** ranging from **0 TO 9 , BOTH INCLUSIVE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric
keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

1. Let $\int_0^a \frac{\cot^{-1}\left(e^x\right)+\cot^{-1}\left(e^{-x}\right)}{\tan^{-1}a+\tan^{-1}x}\cdot\frac{dx}{x^2+1}=\frac{\pi}{p}\ln q,(a>0)$ where $p, q\in N$, then find the minimum value of $(p+q)$.

2. Let z_1 and z_2 be two complex numbers such that $|z_2|=1$ and $\frac{z_1-1}{z_1+1}=\left(\frac{z_2-1}{z_2+1}\right)^2$, then find the value of $|z_1-1|+|z_1+1|$.

3. Let $f(x)$ be a polynomial satisfies the relation $f\left(f\left(f(x)\right)\right)+(1-p)f(x)=3,\ \forall x\in R$, where p is any real number. If leading coefficient of $f(x)$ is 2, then find the values of $\left(\frac{d}{dx}\left(f\left(f(x)\right)\right)\text{at}\,x=p\right)+\int_{-p}^{p}\left(2f^{-1}(x)+1\right)dx$.

4. Let S be the set of all determinants such that the product of elements of any row or column is -1 and all entries belongs to the set $\{-1,1\}$. For example $\begin{vmatrix}1&-1&1\\1&1&-1\\-1&1&1\end{vmatrix}$ is an elements of S and number of elements in S is m.

   Let $P=\begin{bmatrix}3&-2&3\\2&-2&3\\0&-1&1\end{bmatrix}$ and trace of the matrix $adj\left(adj\,P\right)$ is n then find the value of $\frac{m}{n}$.

5. A line passes through the point $P(2, 3)$ and makes an angle of $30°$ with the positive direction of x-axis. It meets the lines represented by $x^2-2xy-y^2=0$ at the points A and B such that $PA\cdot PB=a$. Find the value of $\left[\frac{a\left(\sqrt{3}-1\right)}{5}\right]$ ([.] denotes greatest integer function).

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6. Given two intersecting circles C_1 and C_2 with centres at M and N respectively. From one of the point of intersection A, tangents are drawn to two circle meeting the circles at B and C respectively. Let P be any point located such that $PMAN$ is a parallelogram and if $AB+BC+CA=k\left(PA\sin A+PB\sin B+PC\sin C\right), k\in R,$ then find k. (where angles A,B,C are angle of ΔABC)

7. If a, b, c, p, q and $r\in R-\{0\}$ such that $ap+bq+cr+\sqrt{\left(a^2+b^2+c^2\right)\left(p^2+q^2+r^2\right)}=0$ then find the value of $\frac{aq}{bp}+\frac{br}{cq}+\frac{cp}{ar}$.

8. A circle passes through the point (3, 4) and cuts the circle $x^2+y^2=a^2$ orthogonally. The locus of its centre is a straight line. If the distance of the straight line from the origin is 3, then a^2 is equal to _____.

SECTION - 2 (Maximum Marks : 24)

This section contains SIX (06) questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

9. All chords of the curve $3x^2-y^2-2x+4y=0$ which subtend a right angle at the origin pass through

A) Centre of the rectangular hyperbola $x^2-y^2-2x-4y=12$

B) The point of intersection of the lines $y+2x=0$ and $x=1$

C) The vertex of the parabola $x^2-2x-4y-7=0$

D) Centre of the circle $x^2+y^2+2x-4y-4=0$

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10. For the 3 events A, B and C, P (atleast one occurring) $=\frac{3}{4}$, P (atleast two occurring) $=\frac{1}{2}$ and P (exactly two occurring) $=\frac{2}{5}$. Which of the following relations is/are CORRECT?

[Note: $P(X)=$probability of occurring of event X]

A) $P(A\cap B\cap C)=\frac{1}{10}$

B) $P(A\cap B)+P(B\cap C)+P(C\cap A)=\frac{7}{5}$

C) $P(A)+P(B)+P(C)=\frac{27}{20}$

D) $P(A\cap \overline{B}\cap \overline{C})+P(\overline{A}\cap B\cap \overline{C})+P(\overline{A}\cap \overline{B}\cap C)=\frac{1}{4}$

11. Let $\vec{a},\vec{b}$ and $\vec{c}$ be three non-coplanar vectors and $\vec{d}$ be a non-zero vector perpendicular to $\vec{a}+\vec{b}+\vec{c}$. Now $\vec{d}=(\vec{a}\times\vec{b})\sin x+(\vec{b}\times\vec{c})\cos y+2(\vec{c}\times\vec{a})$, then

A) $\frac{\vec{d}\cdot(\vec{a}+\vec{c})}{[\vec{a}\,\vec{b}\,\vec{c}]}=2$

B) $\frac{\vec{d}\cdot(\vec{a}+\vec{c})}{[\vec{a}\,\vec{b}\,\vec{c}]}=-2$

C) minimum value of $x^2+y^2=\frac{\pi^2}{4}$

D) minimum value of $x^2+y^2=\frac{5\pi^2}{4}$

12. If $y=f(x)$ is a solution of $\frac{dy}{d(x^2)}+\frac{y}{x^2}=\frac{\sin^{-1}x}{2x^2}$ and $f(0)=0$, then

A) $f(1)=\frac{\pi}{8}$

B) $f(-1)=-\frac{\pi}{8}$

C) $f\left(\frac{1}{2}\right)=\frac{\sqrt{3}}{4}-\frac{\pi}{6}$

D) $x^2f(x)=\frac{1}{2}(2x^2-1)\sin^{-1}+\frac{1}{4}x\sqrt{1-x^2}+\frac{1}{4}$

13. If $\cos\frac{2\pi}{2019}.\cos\frac{4\pi}{2019}.\cos\frac{6\pi}{2019}...\cos\frac{2018\pi}{2019}$ is 2^{-k} where $k=\frac{p}{q}$, $p,q\in N$ and p and q are relatively prime, then

A) $(p+q)$ is a 4-digit number

B) sum of the digits in $(p+q)$ is 2

C) $(p+q)$ is a 3-digit number

D) number of divisors of p is 8

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14. Let $\int_0^\infty e^{-x^2}dx = a$ and $I_n = \int_0^\infty x^n e^{-x^2}dx,\ n \in N$, then

A) $I_n = \left(\frac{n-1}{2}\right) I_{n-2}$

B) $I_n = \frac{(n-1)!}{2^{n-1}\left(\frac{n-2}{2}\right)!}a$, if n is even

C) $I_n = \frac{1}{2}\left(\frac{n-2}{2}\right)!\,a$, if n is even

D) $I_n = \frac{(n)!}{2^{n-1}\left(\frac{n-2}{2}\right)!}$, if n is even

SECTION - 3 (Maximum Marks : 12)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

15. If $\int_0^{4\pi} \ln\left|13\sin x + 3\sqrt{3}\cos x\right| dx = k\pi \ln 7$, then the value of k is:

A) 2 B) 4 C) 8 D) 16

16. If the variable line $y = kx + 2h$ is a tangent to an ellipse $2x^2 + 3y^2 = 6$ then locus of $P(h, k)$ is a conic C whose eccentricity equals:

A) $\frac{\sqrt{5}}{2}$ B) $\frac{\sqrt{7}}{3}$ C) $\sqrt{\frac{7}{3}}$ D) $\sqrt{2}$

17. If complex number z_1 satisfies $\arg z = \frac{\pi}{4}$ and $|z-2-i| = 1$ and z_2 is the complex number which lies on the curve $|z-25i| = 15$ & having least positive argument then minimum value of $|z_1 - z_2|$ will be:

A) $\sqrt{346}$ B) $3\sqrt{74}$ C) $2\sqrt{74}$ D) $\sqrt{276}$

18. If p, q, r are prime numbers and α, β, γ are positive integers such that L.C.M. of α, β, γ is p^3q^2r and greatest common divisor of α, β, γ is pqr, then the number of possible triplets (α, β, γ) will be:

A) 48 B) 36 C) 72 D) 96

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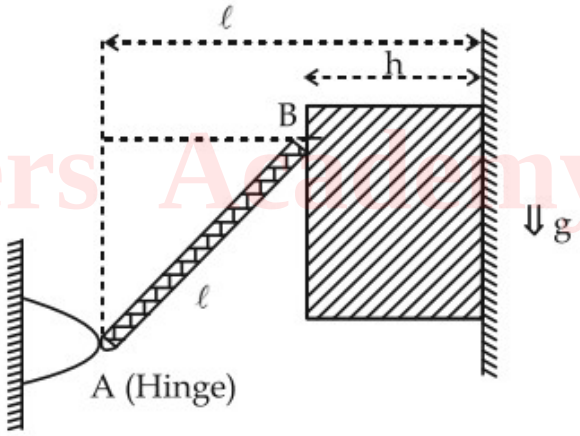
PHYSICS **MAX.MARKS: 60**

SECTION – 1 (Maximum Marks: 24)

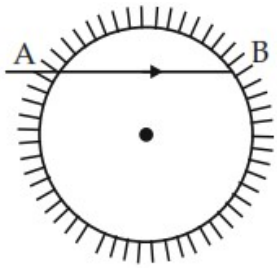
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Negative Marks: -1 In all other cases.

19. The energy of a system as a function of time t is given as $E(t) = A^2 e^{-\alpha t}$, where $\alpha = 0.2\ s^{-1}$. The measurement of A has an error of 1.25%, and the error in the measurement of time is 1.50%. The error in the value of $E(t)$ at $t = 5\ s$ is _____ %.

20. A rectangular block of mass $m = 0.3\ kg$ is held stationary with the help of massless rod of length $\ell = 10\ m$ in vertical plane as shown in the figure. Vertical wall is frictionless and coefficient of friction between block and rod is $\mu = 0.75$. Find the maximum value of h in meter so that block remains stationary. [$g = 10\ m \cdot s^{-2}$]

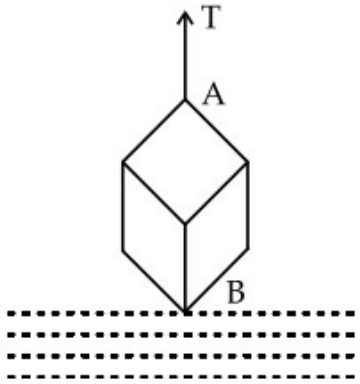


21. A light ray enters horizontally into a vertical cylindrical mirror through a small hole at A. The ray is initially travelling along a chord AB whose length is $\left(\frac{\sqrt{5}+1}{2}\right)R$, where R is the radius of the cylinder. After how many reflections on the inner wall of the mirror, the light ray will be incident again at point A? Given: $\cos 36^\circ = \frac{\sqrt{5}+1}{4}$

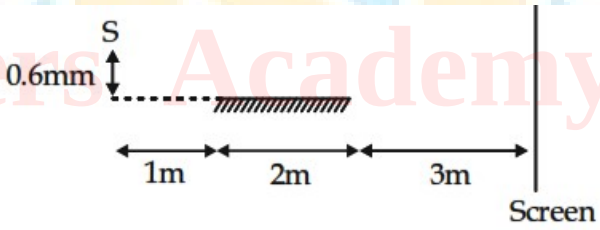


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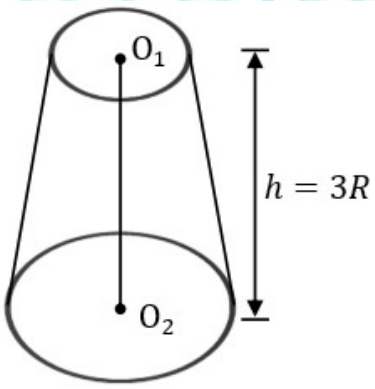
22. A cube of volume $1\ m^3$ and specific gravity 2 is very slowly being lowered into a lake with help of a massless string tied to one of its vertices as shown. Initially the vertex B is just touching the surface of water and finally, vertex A is just beneath the surface of water. For the process of immersion, answer the following questions (Ignore change in KE of water). Work done by gravity on water + cube system is $\sqrt{\frac{3}{2}X}\times10^4$ Joule. Find the value of X. $[g=10\ m\cdot s^{-2}]$



23. In a particular set up, interference is observed on a screen as shown. Wavelength of light emitted by source is $4000\,\text{Å}$. Out of two coherent sources, one is obtained by reflection with a plane mirror of length 2 *m*. Find the number of points on the screen where intensity is minimum.



24. A frustum is cut from a right circular cone. The two circular faces have radii R and $2R$ and their centers are at O_1 and O_2 respectively. Height of the frustum is $h=3R$. A point charge having a charge Q is placed at the midpoint of the line joining the centers. Flux of electric field through the curved surface is $\frac{3Q}{2\varepsilon_0}\left(\frac{1}{\sqrt{a}}+\frac{1}{b}\right)$. Both a and b are integers. Find the value of $|a-b|$.



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25. A radioactive sample can decay by either of two processes X and Y. The half-life for process X is $\frac{1}{4}hours$ and for process Y is $\frac{1}{6}hours$. In a hypothetical process the sample decays for first half hour by process X, for next one hour by process Y and for next half hour by processes X and Y both. The initial number of nuclei of the sample is N_0. After $2\,hours$, the number of undecayed nuclei of the sample are $\frac{N_0}{32}\left(\frac{1}{2}\right)^n$. Find the value of n.

26. A particle is moving in a circle of radius R with constant speed. Time period of the particle is $T = 2.30\ s$. In time $t = T/6$, if the difference between average speed and average velocity of the particle is $2\ ms^{-1}$. The radius of the circle is $R\ m$. Find the value of $\sqrt{R}$.

SECTION - 2 (Maximum Marks : 24)

This section contains SIX (06) questions.
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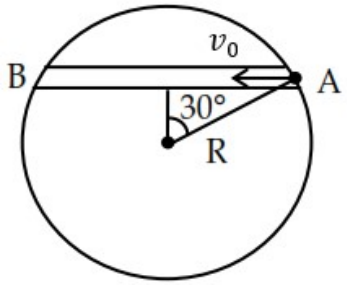
27. Two particles of same mass m are connected by an ideal string of length ℓ. Consider system to be present in free space and string in just taut. Now an impulse is imparted to particle A in direction perpendicular to string. Due to this, instantaneously it starts moving with speed u. In subsequent motion.



A) Magnitude of tension in string remains constant.
B) In a general interval of time $t = t$ to $t = t + T$, the particle B travels a distance equal to 4ℓ.
C) In the reference frame fixed to ground consider initial line along AB. The particle B performs SHM along this line.
D) In the frame of reference fixed to centre of mass of particles, the particle B performs periodic motion with period $T = \frac{2\pi\ell}{u}$.

For More Join: @IITJEE_Advanced

28. A smooth tunnel is dug across the earth along a chord AB as shown. A particle is projected inside the tunnel from end A with an initial velocity $v_0 = \frac{\sqrt{3\,g_0 R}}{2}$. Choose the correct option(s). [$g_0 =$ acceleration due to gravity on the surface of Earth, R = radius of earth]



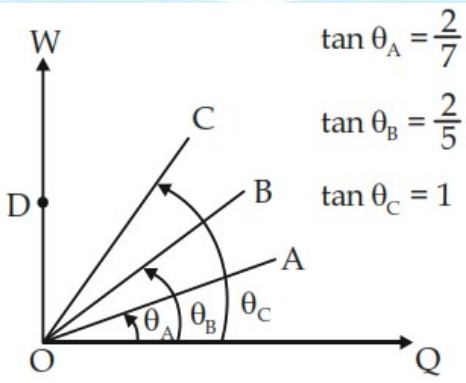
A) The time taken by the particle to move from end A to other end B of the tunnel is $\frac{\pi}{3}\sqrt{\frac{R}{g_0}}$

B) The time taken by the particle to move from end A to other end B of the tunnel is $\frac{2\pi}{3}\sqrt{\frac{R}{g_0}}$

C) The maximum speed of the particle inside the tunnel is $\sqrt{g_0 R}$

D) The maximum speed of the particle inside the tunnel is $\sqrt{2\,g_0 R}$

29. Figure below shows four lines OA, OB, OC, OD (representing various processes such as isothermal, adiabatic, isobaric for a monoatomic and a diatomic ideal gas) on a work done by the gas (W) versus heat supplied (Q) graph. The initial states of both the gases are the same. The scales for the two axes are same.



A) OA represents isobaric process for diatomic gas

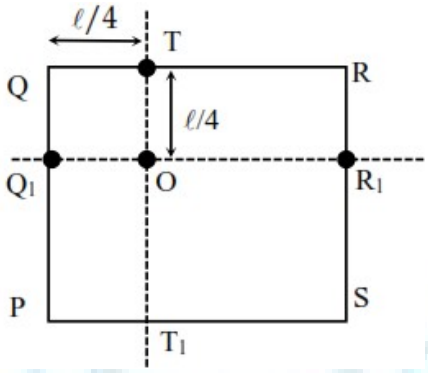
B) OB represents isobaric process for diatomic gas

C) OC may represent isothermal process for diatomic gas

D) OD may represent adiabatic process for diatomic gas

For More Join: @IITJEE_Advanced

30. PQRS is a square of side ℓ. A long wire carrying current i is passing through point O and is normal to the plane of the square. Then choose the correct option(s).



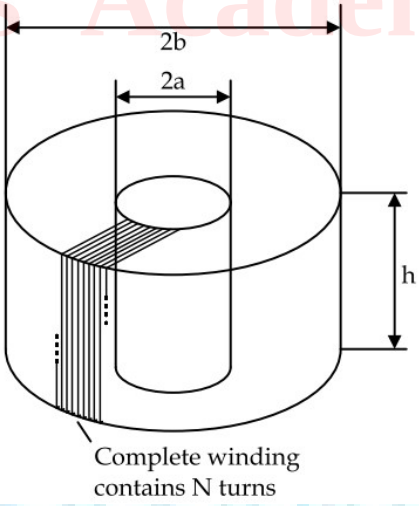
A) $\int_{QRR_1}\vec{B}\cdot\vec{d}\ell=\frac{5\mu_0 i}{4}$, if current in the wire is coming out of page

B) $\int_{QPT_1}\vec{B}\cdot\vec{d}\ell\neq\int_{QRR_1}\vec{B}\cdot\vec{d}\ell$

C) $\int_{Q_1PS}\vec{B}\cdot\overrightarrow{d\ell}=\frac{\mu_0 i}{8}$, if current in the wire is coming out of page

D) $\int_{QRR_1}\vec{B}\cdot\vec{d}\ell=\int_{TRS}\vec{B}\cdot\vec{d}\ell$, for any direction of current in the wire

31. A rectangular toroidal coil has a height h, inner radius a, outer radius b, and N tightly wound turns carrying current I as shown. Assume the magnetic field inside is purely azimuthal and zero outside. Which of the following statements is/are correct?



A) The magnetic field at a radial distance r $(a<r<b)$ from the center is $B(r)=\frac{\mu_0 NI}{2\pi r}$

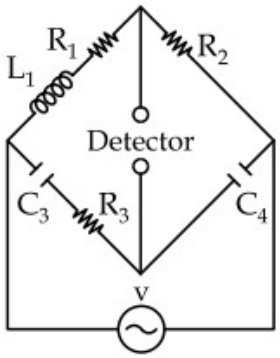
B) The total magnetic flux linked with the coil is $\Phi=\frac{\mu_0 NIh}{2\pi}ln\left(\frac{b}{a}\right)$

C) The self-inductance of the coil increases if the inner radius a decreases

D) The self-inductance of the toroidal coil is $\frac{\mu_0 N^2 h}{2\pi}ln\left(\frac{b}{a}\right)$

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32. A bridge for measuring the inductance L_1 & resistance R_1 of a coil is shown in figure. R_2 & R_3 are resistors C_3 & C_4 are capacitors. For bridge to be balanced i.e. the detector always reads zero, choose the CORRECT option(s).



A) $\frac{L_1}{C_4} = R_2R_3$ B) $L_1 = \frac{C_4R_3}{R_2}$ C) $R_1 = \frac{R_2C_4}{C_3}$ D) $L_1 = \frac{C_4R_2}{R_3}$

SECTION - 3 (Maximum Marks : 12)

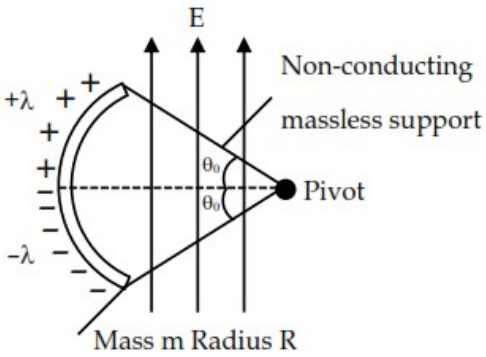
This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

33. A girl is playing with a ball of yarn with a radius $R = 5$ cm, which is formed by a $\ell = 100$ m long string. She stands on a very rough inclined plane with a slope angle $\alpha = 3^\circ$. She kicks the ball along the slope upwards, but one end of the string stays stuck at the point where she kicked it away, so the string is unraveling as the ball travels up the slope until the whole ball unravels and only the straight string remains on the slope. Find the smallest initial velocity of the ball needed to reach the state described above.
$[g = 10\ m \cdot s^{-2}]$

A) 3 ms^{-1} B) 6 ms^{-1} C) 9 ms^{-1} D) 12 ms^{-1}

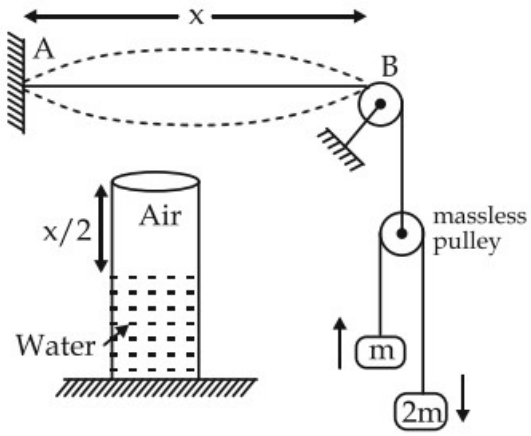
34. Figure shows a charged rod of mass m, bent in the form of an arc of a circle of radius R. The charge distribution on the rod is shown in figure. The assembly is kept in a uniform electric field as shown. For small angular displacement of the system about the pivot, time period of oscillation is: (Neglect gravity)



A) $2\pi\sqrt{\frac{m}{\lambda \sin^2\frac{\theta_0}{2}E}}$ B) $2\pi\sqrt{\frac{m}{2\lambda\sin^2\theta_0 E}}$ C) $\pi\sqrt{\frac{m}{\lambda\sin^2\theta_0 E}}$ D) $\pi\sqrt{\frac{m}{\lambda\sin^2\left(\frac{\theta_0}{2}\right)E}}$

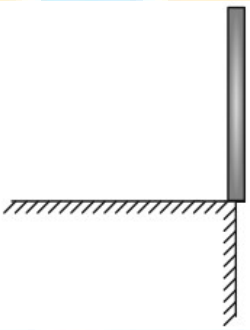
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35. AB wire is vibrating in its fundamental mode. Wire AB is in resonance with resonance tube in which air column is also vibrating with its fundamental mode. Sound speed is $400\ m/s$ and linear mass density of AB wire is $10^{-4}\ kg/m$ and $g = 10\ m/s^2$. Find the mass m. Neglect the masses of wires in comparison to mass of the blocks.



A) $0.6kg$ B) $1.2kg$ C) $1.0kg$ D) $2.0kg$

36. A uniform rod is placed on the edge of a table in a vertical position. The edge of table is rough and very sharp. The rod is gently disturbed from rest and it starts tipping rightwards about the edge. When the rod leaves the contact with edge of table, the angle it makes with the vertical is



A) $\sin^{-1}\frac{1}{2}$ B) $\cos^{-1}\frac{2}{3}$ C) $\sin^{-1}\frac{3}{4}$ D) $\sin^{-1}\frac{4}{5}$

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CHEMISTRY **MAX.MARKS: 60**

SECTION – 1 (Maximum Marks: 24)

This section contains SIX (08) questions.
The answer to each question is **A SINGLE DIGIT INTEGER** ranging from **0 TO 9 , BOTH INCLUSIVE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric
keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

37. Oxide (A) of a certain metal contains 22.55% of oxygen by mass. Oxide (A) adopts rock salt structure and is a component of fertilizers and food additives. It forms green crystals. Another oxide (B) of same metal contains 50.48% of oxygen by mass. Oxide (B) is dark green. It has similar structure as pyrosulphate and pyrophosphate. What is the sum of oxidation states of metal ions in oxide (A) and oxide (B)?

38. The total number of complexes among the following which are paramagnetic:

$K_3[Fe(CN)_6], [Cr(H_2O)_6]Cl_3, [Pt(NH_3)_2Cl_2], [Cr(en)_3]Cl_3, [Cr(NH_3)_4Cl_2]Cl,$

$[Cr(H_2O)_5Cl]SO_4, [Co(H_2O)_4Cl_2]NO_3, [Zn(SCN)_4]SO_4$

39. How many of the following do not hydrolyse completely in ordinary conditions?

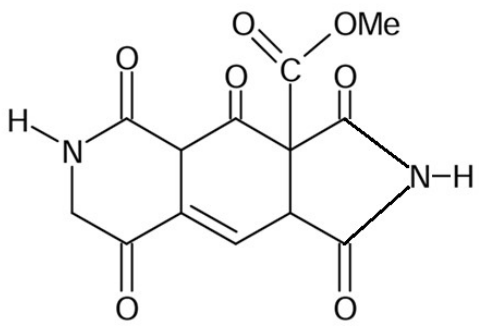
$BF_3, P_4O_{10}, SbCl_3, SiF_4, H_2S_2O_7, Cl_2O_7, H_4P_2O_5, N_2O_3, CO_2, SOCl_2, NCl_3, AsCl_3, SF_6$

40. $\underset{\text{(All isomers)}}{C_2FClBrI} \xrightarrow{H_2/Ni}$ Number of products 'X' (excluding stereoisomers)

$\underset{\text{(All isomers)}}{C_4H_8(\text{alkene})} \xrightarrow{H_2/Ni}$ Number of products 'Y' (excluding stereoisomers)

Find the value of $X+Y$.

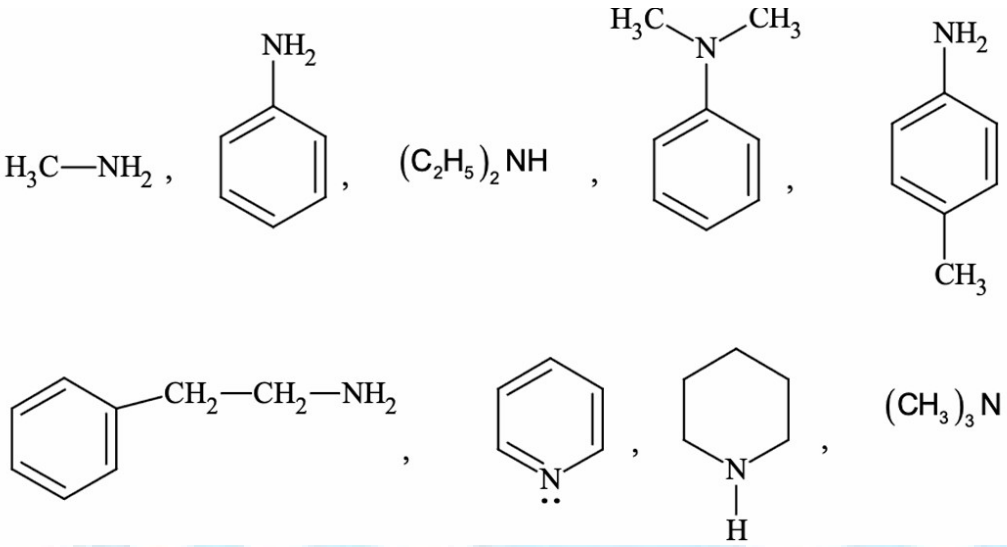
41. Consider the following compound (X):



If 0.5 moles of the compound (X) are hydrolysed in alkaline medium and then heated. How many moles of CO_2 are produced ?

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42. The number of following amines, which are relatively more basic than benzyl amine is/are 'a' and the amines which can be easily prepared by Gabriel's phthalimide synthesis is/are 'b' then the value of $|a-b|$ would be:



43. How many of the following will have no unpaired electrons if Hund's rule is violated in filling of electrons.

$B_2, C_2, N_2, N_2^+, O_2, O_2^-, F_2, CO, NO$

44. 20 g of naphthoic acid $(C_{11}H_8O_2)$ dissolved in 50 g of benzene $\left(K_f = 1.72\ K\ kg\ mol^{-1}\right)$ shows a depression in freezing point of $2K$. How many molecules of naphthoic acid are associating in benzene if the degree of association is 100%.

SECTION - 2 (Maximum Marks : 24)

This section contains SIX (06) questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

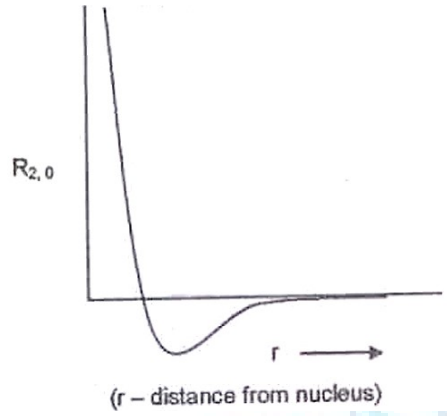
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

45. Choose the correct statements among the following regarding hydrogen atom? (All graphs given below are indicative in nature, not to the scale)

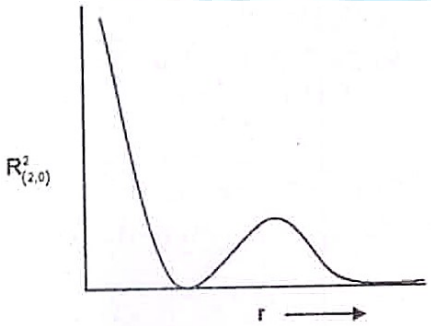
A) Radial wave function $R(r)$ gives information about the energy and size of the orbital.

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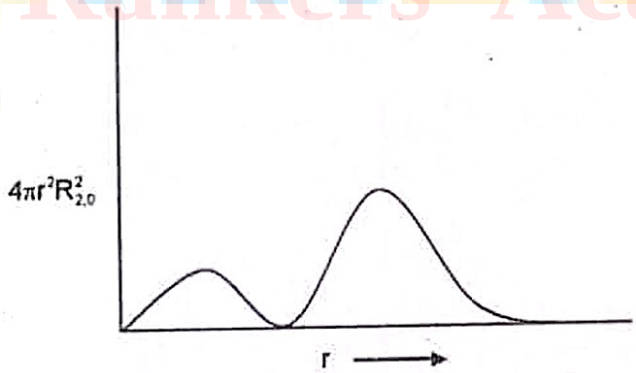
B) Plot of $R_{(2,0)}(r)$ versus r, for $2s$ electron is



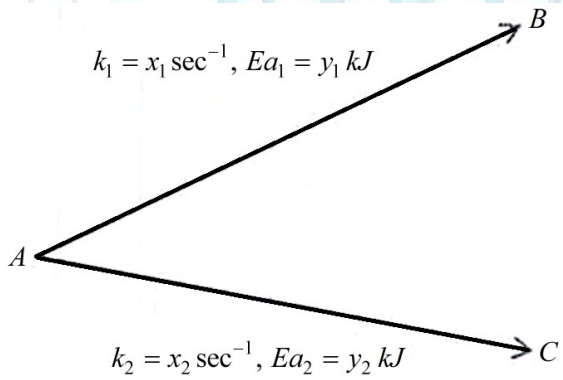
C) Plot of $R^2_{(2,0)}$ versus r for $2s$ electron is



D) Plot of function $4\pi r^2 R^2_{2,0}$ against r (distance form nucleus) is



46.



(Given that $y_1 > y_2$)

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Choose the correct statements regarding above chemical reaction:

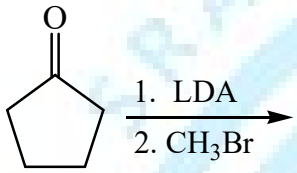
A) Overall rate of reaction is equal to sum of formation of B and C

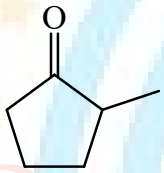
B) Overall activation energy of the reaction $E_a = \frac{x_1y_1 + x_2y_2}{x_1 + x_2}$.

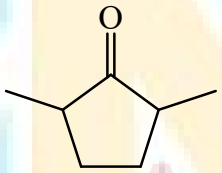
C) Half life of reaction is $\frac{\ln 2}{x_1 + x_2}$.

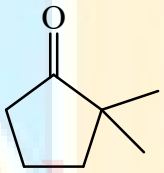
D) As the temperature decreases rate of reaction of both reactions decreases but into of rate of reaction $A \rightarrow C$ decrease more than $A \rightarrow B$

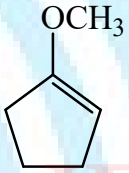
47. The possible product obtained by the following reaction is/are [consider all major and minor products]



A) 

B) 

C) 

D) 

48. Which of the following statement(s) is/are correct for froth floatation process?

A) Pine oil is used as collector.

B) Na_2S acts as activator.

C) Xanthates used as frothers.

D) $NaCN$ used as depressant to separate ZnS from PbS.

49. Select the correct statement(s)

A) An adiabatic system can exchange energy with its surroundings.

B) A thermodynamic property which is intensive is additive.

C) Work done in free expansion is zero.

D) For an isolated system, the entropy can never decrease.

50. Electron affinity of the elements or ions is correctly shown in

A) $S > O^-$ B) $O > S^-$ C) $O^- > S^-$ D) $N^- > S$

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SECTION - 3 (Maximum Marks : 12)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) out of which **ONLY ONE** option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and -1 in all other cases.

51. Aqueous solution of metal ion

$M^{n+} + KI \rightarrow$ Black ppt (N) $\xrightarrow[\text{suspension is heated}]{\text{aqueous}}$ Orange ppt(P). The "P" is

A) $Bi(OH)_3$ B) Hg_2I_2 C) BiI_3 D) $BiOI$

52. In which of the following cells, use of KCl as salt bridge would not change the EMF of the cell

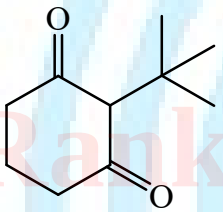
A) $Ag_{(s)}|0.1M\ AgNO_3||0.2M\ AgNO_3|Ag_{(s)}$

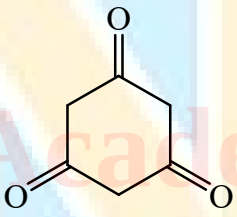
B) $Cu_{(s)}|0.1M\ CuCl_2||0.1M\ AgNO_3|Ag_{(s)}$

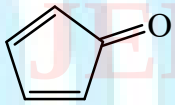
C) $Pt, Cl_{2(1atm)}|HCl(0.5M)||HCl(0.05M)|Cl_{2(1atm)}, Pt$

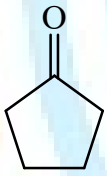
D) $Ag_{(s)}|0.2M\ AgNO_3||HCl(0.2M)|Cl_{2(1atm)}, Pt$

53. Compare the $C-O$ bond lengths in the following ?

(a) 

(b) 

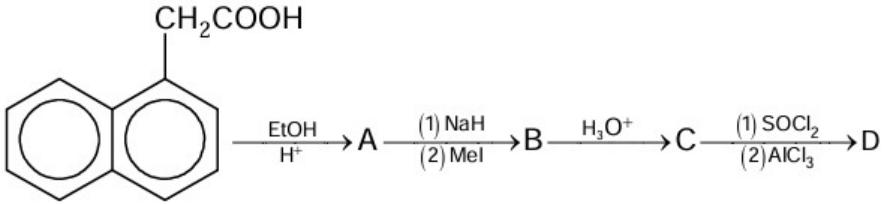
(c) 

(d) 

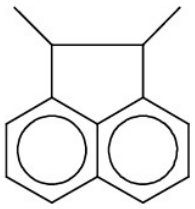
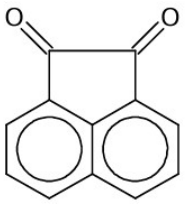
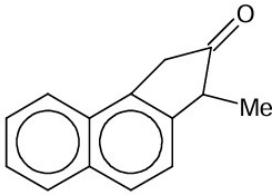
(e) CO_3^{2-} (f) $RCOO^-$

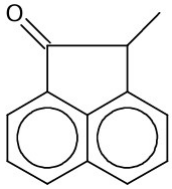
A) b > e > f > c > d > a B) e > b > f > d > c > a

C) b > e > f > d > a > c D) d > c > a > b > f > e

54. 

D is ?

A)  B)  C) 

D) 

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